

Jerry Zhang

📞 832-557-4546 ✉ jerryzhang11@tamu.edu  [linkedin.com/in/jerry-zhirui-zhang](https://www.linkedin.com/in/jerry-zhirui-zhang)

EDUCATION

Texas A&M University

May 2028

B.S. in Mechanical Engineering (Honors); Minor in Japanese

GPA: 3.84

Phillips 66 Community STEM Scholarship Recipient (66/1,000 selected)

SKILLS

Engineering Skills: CAD (Fusion 360 & SolidWorks), Mechanical Design, Prototyping, Testing & Validation, Manufacturing & Fabrication, 3D Printing, Data Analysis, Troubleshooting, Technical Documentation

Programming Skills: Python, Java, C++, R

Languages: Mandarin (Native/Fluent), Japanese (Fluent)

EXPERIENCE

Houston Methodist Research Institute — Visiting Student

Mar–Aug 2023; May–Aug 2026

- Designed and executed multi-variable experiments applying the same controlled-variable, iterative testing methodology used in engineering root-cause analysis of complex biological signaling systems.
- Operated and calibrated precision lab instrumentation (flow cytometry, imaging systems) and analyzed data with statistical methods, ensuring measurement accuracy and reproducibility to publication-grade standards
- First author **publication** in Nature Communications, translating complex technical findings into clear, structured documentation and presenting results at national conferences

Science Olympiad — National Medalist & Engineering Mentor

Sep 2021 – May 2024

- Earned **2nd place nationally (2023)** and **4th place (2022)** in aviation design, plus **3rd place (2022)** in bottle rockets; qualified for Nationals by placing **top 2 of 30 Texas teams**, then competed among **60 teams nationally** at the 2022 Caltech-hosted tournament.
- Mentored 15+ peers in CAD modeling, prototyping, and testing; raised \$5K in funding and managed material budgets while coordinating resources for a 100+ member team.

La Plume Bakery & Café — Bakery Assistant / Acting Manager

Feb 2025 – Aug 2025

- Handled daily operations serving 50+ customers/day and producing 10+ custom cakes weekly, including wedding and specialty desserts.
- Ran full small business operations for 1.5 months overseeing finances, ingredient procurement, staff/vendor coordination, and pricing strategy
- Managed Instagram marketing (photo/video production), growing following by 500+ and increasing customer engagement

SELECTED PROJECTS

F1D Model Airplane Competitions | CAD, Laser Cutting, Flight Testing

Sep 2021 – Sep 2024

- Designed and fabricated a competition airframe via laser-cut balsa fuselage/rib structures and carbon fiber spars with Mylar skinning, balancing structural rigidity against a strict weight budget for flight performance
- Conducted iterative flight testing to optimize angle of attack, pitch trim, and rubber motor density, improving flight efficiency, climb rate, and duration by up to 50%

Gravitationally Powered Model Cars | Fusion 360, 3D printing

Sep 2023 – May 2024

- Designed and fabricated a gravity-powered vehicle, using CAD-modeled, 3D-printed interchangeable components to rapidly test design variables and iterate toward an optimized final build
- Engineered a custom spring-assisted launch system leveraging pulley mechanical advantage, improvising with limited tools and budget to meet aggressive weight and accuracy targets

TEACHING EXPERIENCE

Peer Teacher — Intro to Python @ Texas A&M University

Aug 2026 – Present

- Support 80+ engineering students with Python programming and debugging through weekly office hours and in-class assistance.
- Dedicate 7+ hrs/week to evaluating and grading student work, while guiding students through mathematical and programming approaches to complex problems.